

Wearable Biofeedback for Knee Joint Health

Improving Haptics for Precision Rehabilitation

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ABSTRACT

The human body has the tremendous capacity to learn a new way of walking that can reduce the risk of musculoskeletal disease progression. Wearable haptic biofeedback has been used to teach this new gait to patients with knee osteoarthritis, enabling reductions in pain and improvement in function. However, this promising gait retraining therapy is not yet a part of standard clinical practice. Here, I propose a two-pronged approach to improving the design and implementation of biofeedback for gait retraining. The first section concerns prescription, aiming to provide clinicians with a model that predicts the mechanical outcome of gait retraining in order to best guide their treatment decisions. The second section concerns learning, seeking to understand how internal physiological state and external environmental factors influence the process of learning a therapeutic gait. This work aims to address the challenges keeping this intervention from being widely used to maintain pain-free mobility throughout the lifespan.

CCS CONCEPTS

• **Human-centered computing**; • **Human computer interaction (HCI)**; • **HCI design and evaluation methods**; • **Laboratory experiments**; **Interaction devices**; • **Haptic devices**; **Empirical studies in HCI**;

KEYWORDS

knee osteoarthritis, biofeedback, biomechanics, precision health

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1 INTRODUCTION

Precision health is the future of healthcare. A widespread ability to provide the *right care* to the *right patient* at the *right time* will be catalyzed by recent and accelerating advances in areas such as omics, individualized disease models, and distributed sensing [1]. Wearable sensing devices, including smartwatches, are already becoming a massive source of data and are emerging as a platform for on-the-fly feedback on user state. With the right insights, wearables

could give patients an active role in their treatment and recovery. This method of sensing, synthesizing, and providing feedback on biological signals—beyond what is possible with the organic body—has already been applied to exoskeleton control [2], running form training [3], and mindful meditation [4].

However, in applications relating to musculoskeletal health, we still have limited knowledge of what kind of feedback (haptic, wearable, cumulative?) provides the most effective behavioral change. Precision health can lead to more equitable access to health resources across continents, ages, and incomes. Yet, a lack of coherence in feedback strategies and our still-evolving models of the musculoskeletal system prevent its wide-scale use for mobility-limiting conditions. A smartwatch buzzes the same way to notify you of an email or when you’ve completed your fourth mile; no commercially available wearable can inform you about the health of your joints or how your latest run might affect them over time. In my thesis research, I use the treatment of knee osteoarthritis as a case study for this emerging paradigm in healthcare and outline how my methods and findings extend into domains including haptics, behavioral change, and participatory design.

1.1 My Research and Background

I am a fourth-year Ph.D Candidate in Mechanical Engineering, enrolled in a joint program between Carnegie Mellon University and the Max Planck Institute for Intelligent Systems. I am co-advised by Dr. Eni Halilaj (CMU) and Dr. Katherine J. Kuchenbecker (MPI-IS). After spending the first portion of my degree in Pittsburgh, USA, I moved to Stuttgart, Germany, after submitting my thesis proposal in December 2021. My anticipated thesis defense date will be in May 2024. I have never previously attended a Doctoral Consortium.

1.2 Related Work

Knee osteoarthritis is a disruptive and costly musculoskeletal disease characterized by the degradation of cartilage in the knee joint. At present, there is no pharmacological cure. Pain is managed pharmacologically, while the joint is left to degrade until the patient is ready for a total knee replacement—a period that lasts over a decade on average, with no clinically-accepted disease-modifying intervention in the interim [5]. Because osteoarthritic disease progression is mediated by mechanical loading on the joint, **redistributing** or **reducing knee loading during gait** is expected to slow cartilage breakdown. Unlike other interventions such as knee braces, shoe wedges, or surgical procedures, changing the way a patient walks—also called **gait retraining**—is as effective in redistributing force across the joint as a high tibial osteotomy surgery [6, 7], is easy to learn [8], and results in significant improvements to pain and function over time [9].

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One way to alter knee loading during gait is to walk with the feet pointed slightly inwards (toe-in) or outwards (toe-out) [10]. Changing the orientation of the foot shifts the position of the knee joint in the frontal plane and the location of the ground reaction force underneath the foot, thereby reducing the moment about the knee joint, also called the **knee adduction moment (KAM)**. The KAM is one of the commonly used surrogate measures of knee loading because measuring *in vivo* joint loads requires invasive surgery to instrument the knee joint. Therefore, reducing the peak KAM is the primary target of many gait retraining paradigms [11], although KAM impulse or values derived from musculoskeletal simulations are also used. Previous work has settled on vibrotactile haptic feedback as the optimal method to teach toe-in gait due to its low-profile form factor and ease of use. However, despite over a decade of research and findings that gait retraining with haptic feedback can slow the rate of cartilage degradation [12], gait retraining is not yet part of standard clinical practice.

Although the clinical outcomes are promising, clinicians are not ready to prescribe gait retraining as a treatment for knee osteoarthritis. First, the clinician must be able to estimate the therapeutic benefit of the proposed treatment. No methods currently exist to automate the procedure of predicting KAM reduction in response to gait retraining. For example, customizing the change in toe-in angle to the individual results in a larger peak KAM reduction than a non-custom intervention [13]. However, determining this target toe-in angle is a time-consuming process of acclimation and evaluation that can be performed only in the gait lab. At present, researchers must iteratively evaluate KAM reduction at a range of foot progression angles, relying on force plates and motion capture to compute and compare joint moments. Although proposed methods for estimating KAM using wearable sensors [14, 15] or synthetic video data [16] could help reduce reliance on gait lab equipment, these approaches still require gait data collection, which is infeasible in the doctor's office.

Secondly, individual responses to gait retraining can be highly variable, with some patients failing to learn or retain the new gait [9, 13]. Inconsistent responses could be due to how error is encoded in the haptic device or to individual variability in sensitivity to feedback. Although vibrotactile feedback that “pulls” the foot in the correct direction—e.g., a vibration on the inside of the foot to indicate that the next step should be pointed in more—is preferred for gait retraining [17], past experimental paradigms have not employed feedback based on error magnitude. Knowledge of error magnitude is considered necessary for optimal motor learning [18]. However, even with an informative error signal, patients can vary dramatically in their physiological states. For instance, skin sensitivity to vibration can influence the perceived strength of a cue [19], and a lower proprioceptive ability is correlated to slower motor learning [20]. These findings from adjacent fields suggest that biofeedback for gait retraining could benefit from incorporating scaled error into the feedback system and compensating for the effect of individual differences in vibration sensitivity and proprioception. The role these factors play in effecting clinically-relevant changes in gait is not yet known.

2 PROPOSED AND COMPLETED WORK

This work combines clinical musculoskeletal biomechanics and haptic intelligence (Figure 1). The first section of this research description details a modeling project in biomechanics, with an eye toward the clinician experience and their need for predictive model transparency (2.1). The second section brings principles of user interaction, physiology, and wearable haptic design (2.2) into the biomechanical health space.

2.1 Explainable Precision Rehabilitation

A custom gait retraining plan results in a better therapeutic response [13] but requires gait laboratory instrumentation not typically available to clinicians. Here, I aim to predict the extent to which a patient with knee osteoarthritis will benefit from a non-invasive gait retraining therapy using only measures that can be easily collected in the clinic. The linear model used in the prediction provides a clear relationship between therapeutic outcomes and patient-specific features. With this model, clinicians can use data collected in the clinic to easily identify patients who would respond to therapeutic gait retraining and compare their expected benefit to predicted surgical outcomes to determine an optimal treatment path.

2.1.1 Proposed Research in Explainable Precision Rehabilitation.

- **Goal 1:** Predict the extent of KAM reduction using features that can be easily collected in a clinical setting.
- **Goal 2:** Evaluate models to predict KAM reduction in a population of patients with knee osteoarthritis.

To translate the prescription of gait retraining to the clinic, I built a regression model that uses minimal clinical data—a set of six features easily obtained in the clinic—to predict the extent of KAM reduction after toe-in gait retraining [21]. For such a model to generalize, the training data must be large and variable. However, sharing data is not yet common in biomechanics, and large public datasets that contain different gaits for the same patient are time-consuming and costly to collect. Therefore, I generated this dataset synthetically. First, I used a ground-truth dataset with both baseline and toe-in gait trials ($N = 12$) to learn how gait changes with toe-in and applied these gait changes to a large ($N = 138$) dataset of healthy participants to create the model's training set. I then tested the trained regression model on data collected by a separate research group ($N = 15$). The KAM reduction was predicted with a mean absolute error of 0.134%, scaled to body weight and height (%BW*HT). This error was smaller than the standard deviation of the first peak KAM during baseline walking averaged across test subjects (0.306%BW*HT) or within the range of how much their KAM varies when walking normally. The data, models and code necessary to reproduce the results are available on GitHub [21].

Although this study was the first to automate the procedure of predicting KAM reduction after a gait retraining, a significant limitation was that the model was only tested on healthy control subjects without knee osteoarthritis. In response to this need, I am collaborating with researchers at the University of British Columbia and Stanford University, who have collected biomechanical data of patients with knee osteoarthritis walking over an in-ground force plate and on an instrumented treadmill at varying toe-in

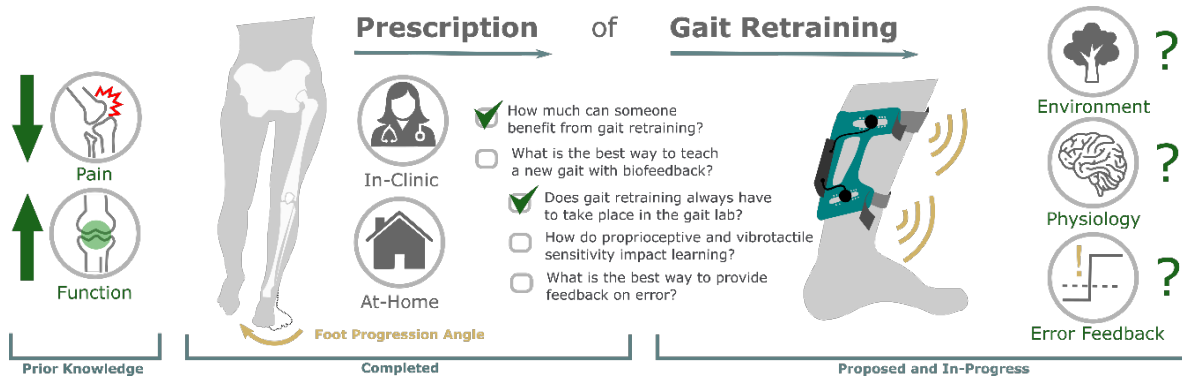


Figure 1: Gait retraining, although linked to positive clinical outcomes, requires more study before it can be prescribed.

and toe-out angles. Together, we seek to evaluate this model on patient data and determine how the model could be improved by synthesizing training data from a more diverse starting dataset. Our insights will guide the future direction of predictive models based on synthetic data, helping to create new approaches to individualized rehabilitation.

2.1.2 Broader Impacts of Explainable Precision Rehabilitation. Although this research is predominantly biomechanical, it is firmly grounded in principles of human-computer interaction. The primary motivation behind this work was to create a clinician-facing tool that acts as an interface between the human patient and the model that predicts their rehabilitation outcome. Unlike other projects in biomechanics that seek to prescribe an intervention, our focus was to give the clinician a deeper insight into the expected outcome of an intervention so that they can be better informed about possible treatment paths for their patient. Validating the model on various datasets from a diversity of laboratories will build trust in the system and improve the reliability of the model; the coefficients of the regression model are linearly interpretable, allowing the prediction to be interrogated for sensitivity to each predictor. Finally, synthetic data generation is a critical step toward increasing the diversity of biomechanical data. Often, predictive models are trained on whoever is available in the researcher’s lab, resulting in models that are trained to perform well on young, healthy males but are brittle when exposed to patient data.

The next step after creating these models and evaluating them on the target patient populations will be to conduct user studies or structured interviews with the end-user: the clinicians responsible for planning a patient’s treatment. Ultimately, including the clinician in the development and deployment of the model will help ensure translation between the research laboratory and clinical practice. Future user studies and structured interviews will shed light on clinician expectations for precision health devices for at-home rehabilitation.

2.2 The Human in the Haptic Feedback Loop

Despite best efforts to assert the contrary, a human user is not a deterministic system. We have desires, goals, discomforts, abilities, weaknesses, and predilections to caffeine or insomnia. However, this complexity is sometimes treated as an inconvenient truth of

human subject studies, with the hope that it will not affect analysis if the number of study participants is high enough. Here, I propose to study how physiological and environmental states impact the process of learning a therapeutic gait as a first step in understanding the heterogeneity of responses to gait retraining. Whereas Explainable Precision Rehabilitation (2.1) simulated the ideal experimental outcome of retraining—given only a patient’s mechanics, what is their best possible reduction in KAM?—The Human in the Haptic Feedback Loop (2.2) answers the question of what lifestyle or physiological factors impede or improve a patient’s ability to achieve that ideal reduction.

2.2.1 Proposed Research in the Human in the Haptic Feedback Loop.

- **Goal 1:** Examine whether a wearable haptic device can be used to learn a new gait in a natural environment.
- **Goal 2:** Identify if lifestyle and physiological attributes impact the learning of a new gait.
- **Goal 3:** Determine if knowledge of error results in better learning and retention versus unscaled feedback.

Despite the growing promise of wearables for remote health monitoring and intervention, research on the benefits of technology-assisted gait rehabilitation is still limited to laboratory settings, which might not realistically represent natural environments. I sought to evaluate whether a wearable biofeedback device based on inertial sensing can support learning a new gait in a natural environment to the same extent that it can be learned in a controlled laboratory setting. Using a commercially-available device for gait retraining [22], I taught 30 subjects a toe-in gait: one-third of the subjects trained with vibrotactile feedback on a lab treadmill for 20 minutes, one-third trained the same way but in an outdoor environment, and the remaining third did not receive any training or feedback at all. When comparing the tracking accuracy of the inertial sensor to optical motion capture, I found that participants with smaller feet had less accurate tracking of their toe-in angle: as the algorithm for step detection is based on gyroscopic readings from a sensor affixed to the top of the foot, a smaller foot length may have resulted in smaller signal peaks, more missed steps, and inconsistent haptic feedback. Ultimately, a higher tracking error predicted whether someone would learn the new toe-in gait, whereas the environment they trained in did not. These findings demand further support for kinematic estimation algorithms built for all body

shapes and sizes for wearable gait rehabilitation to move towards unsupervised at-home prescription.

The design of this study could only provide evidence of differences in learning responses across participants. Despite unreliable feedback, most participants in this study were nevertheless able to maintain a toe-in gait. It was unclear if these more adept learners had some intrinsic abilities that enabled them to learn the new gait—a heightened sense of proprioception, better athletic ability, or a lower cognitive load, for instance. In my subsequent study, I aim to collect participant data on physiological state using an Empatica watch [23], which measures electrodermal activity, heart rate, sleep quality, and activity. I will measure individual proprioceptive ability and vibrotactile sensitivity with psychophysical assessments. Using the wearable haptic device I am creating to integrate with optical motion capture, I will study how these physiological values impact how a participant learns and retains a toe-in gait. Based on motor learning theory, I hypothesize that individuals with a higher cognitive load, indicated by higher electrodermal activity, will be more responsive to the feedback and therefore have a steeper learning curve over time. Similarly, I hypothesize that greater proprioceptive and vibrotactile sensitivity will increase accuracy in maintaining the toe-in gait.

A parallel component of what makes biofeedback intelligible is encoding of vibrotactile output. Although wearables for health monitoring have settled on vibrotactile feedback as their preferred medium of communication, present versions hardly take advantage of the myriad ways haptics has advanced to create informative, comfortable, and beautiful devices. Previously-developed devices use “bang-bang” feedback with a bandpass of allowable kinematic angles and a short vibration to indicate the presence of an error. When using this binary feedback method for gait retraining, a past study participant described themselves as “an underdamped oscillator,” unable to scale their next step appropriately. I propose to examine how learning compares between individuals receiving haptic feedback that is scaled by the distance of their toe-in angle from the target and those training with standard feedback. I will also compare individual preference for feedback type with surveys and unstructured interviews to assess trust and comfort with the device.

2.2.2 Broader Impacts of the Human in the Haptic Feedback Loop. A better understanding of how physiology impacts learning with biofeedback will be valuable to both biomechanists and hapticians. For instance, both vibrotactile and proprioceptive sensitivity tend to decrease with age [24, 25]. Biofeedback for rehabilitation, including after stroke or limb loss, may need to account for this variability in older populations. Furthermore, physiological variables that can be estimated with wearable devices could inform interventions appropriate to the individual user and contextualized with their behavioral change. Appropriately cued haptic feedback can contribute to guided learning strategies in haptics, such as in robotic surgical training. In the future, it may be possible to modulate the delivery of feedback based on physiology to optimize learning or compliance; we may find that after a restless night of sleep, less frequent error notifications result in a more successful remote physical therapy session.

In the future, I aim to understand physiology’s impact on learning to improve how wearable devices respond to the body’s rhythms and fluctuations to provide more precise feedback. I will then combine my past work in developing feedback systems for lower limb amputees [26] with advances in textile-embedded sensors [27] to create the next generation of wearables for health monitoring. My long-term goal will be to fuse these sensing and feedback systems with individualized musculoskeletal modeling to robustly track muscle, tendon, and ligament behavior, thereby better informing haptic feedback interventions.

3 CONCLUSION

My research aims are to advance the development of wearable haptics for precision treatment of musculoskeletal disease. My proposed thesis research tackles the open challenges in implementing biofeedback for rehabilitation: prescription, individual variability in learning, and optimal feedback delivery methods. I accomplish this through several technical approaches: I describe a novel computational method to predicting clinical outcomes of a targeted gait retraining intervention; I propose an integrated physiological study to identify the aspects of human behavior that influence learning; I describe a mixed-methods study aimed at contrasting different strategies for haptic feedback to determine how it can be most beneficial for the example application of gait retraining.

Precision health is a forum with many stakeholders. As this paradigm of healthcare emerges, a foundational understanding of its functional components will enable me to target my technical most effectively. In this proposal, I describe my aims to progress the field from an academic and clinical perspective. Still, I recognize that future success entails collaborations with leaders in government, industry representatives, and entrepreneurs to consider the legal, safety, privacy, manufacturing and commercialization steps necessary.

Ultimately, a federated approach to healthcare makes treatment more equitable. Increased access to healthcare resources is a critical priority for me as a native Ukrainian who has seen firsthand what rural and underdeveloped healthcare looks like, even in times of relative peace. Telehealth has become instrumental to many in the past few years; the gradual acceptance of remote diagnosis and treatment gives me hope that, in the future, musculoskeletal disease can be treated in the same way. Improving our understanding of what makes biofeedback most useful is only the first step toward making distributed precision treatment a modern reality.

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